Sampling, measurement and ring-width series

Lesson 4 · 90 minutes · From samples and records to comparable series

Learning objectives

Plan sampling and records

Explain a sampling plan and the metadata it needs.

Measure widths from boundaries

Check the path, spatial calibration, boundary positions and units.

Combine ring-width series

Distinguish detrending, tree weighting and sample depth.

From a specimen to a chronology

01 Question and sampling plan

02 Specimens, images and records

03 Observe and measure boundaries

04 Crossdating and quality control

05 Detrending and chronology

Return to the wood and records when a mismatch appears.

Start sampling with the question

Climate

Choose species, sites and comparison areas relevant to the climate variable.

Forest history

Plan how spatial sampling will capture disturbance and regeneration.

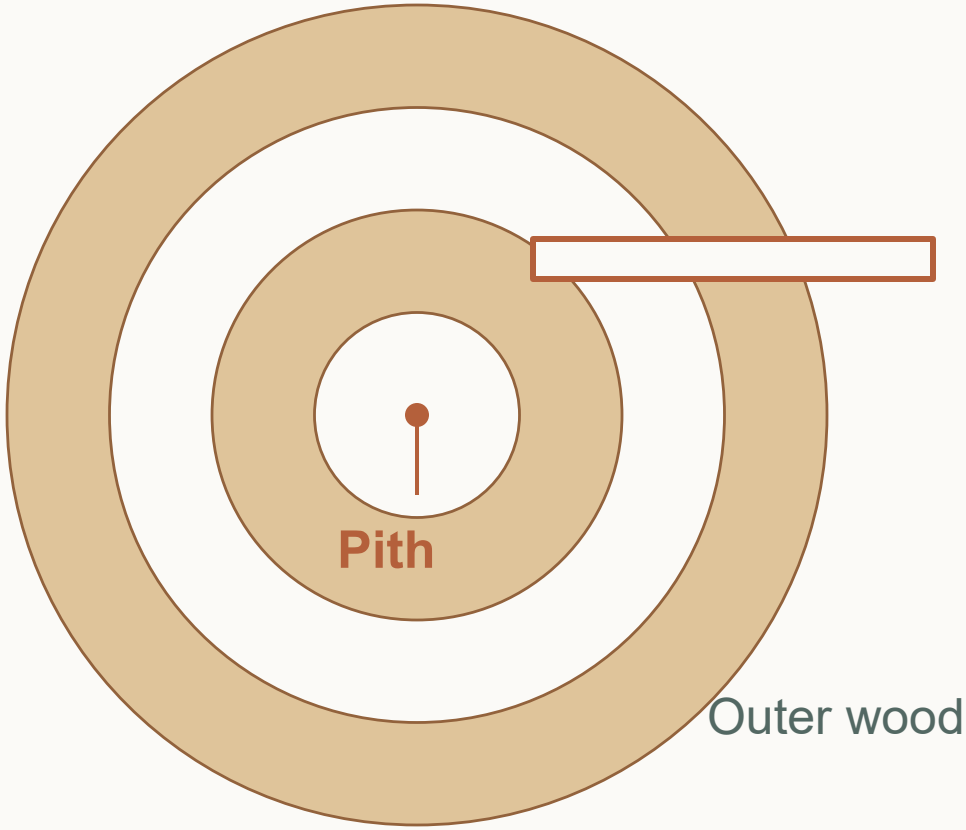
Buildings and objects

Record member location, working or reuse, and surviving inner and outer wood.

What material will be available?

Increment core	**One radial record; document sampling position, losses and injuries.**
Disc or block	Allows comparison across radii and scars; define the sampled extent.
Surface image	Records the readable surface; requires a scale and imaging conditions.

What a core does and does not record



Did the core reach the pith?

Is the outermost ring preserved?

Could bending, breakage or compression affect the record?

Synthetic teaching data / schematic example, not empirical tree measurements.

Link measurements to specimen metadata

Identification

Link site, tree, specimen and path IDs to images and containers.

Collection and condition

Date, species, member position, height, direction, pith, bark and losses.

Processing and measurement

Preparation, image calibration, operator, boundaries, revisions and reasons.

Mini exercise: plan a collection

You want to study moisture variability in a forest with ridges and valleys.

Choose sites, trees, radii and the records to keep.

State one limitation arising from the places you did not sample.

Preserve order during transport and storage

Identify immediately

Link specimen and container IDs; retain orientation and fragment order.

Handle for the condition

Record cracks, decay and moisture; minimise distortion and mixing.

Record the initial state

Save photographs and collection records, including missing material.

Prepare a surface with readable boundaries

Mount and orient

Support the core after checking its orientation and cellular structure.

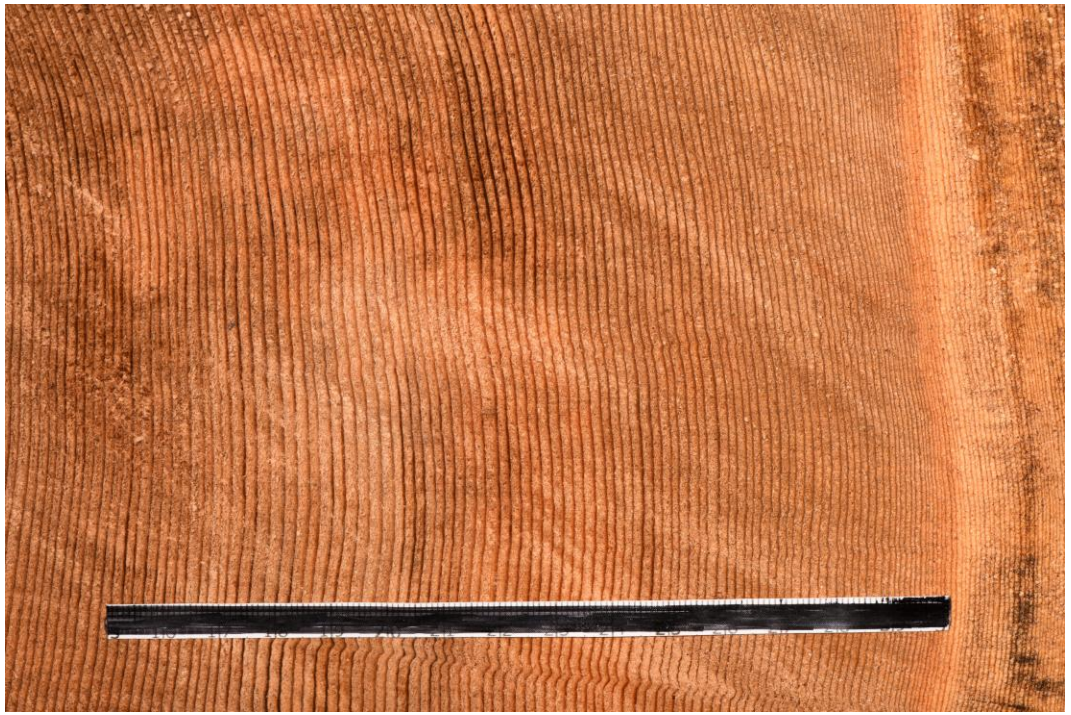
Prepare the surface

Use preparation suited to the material to reveal boundaries and cells.

Inspect the result

Distinguish scratches and preparation marks from narrow increments.

Imaging a wood cross-section



Example of a wood surface. Photo: Hitoshi Yonenobu

Distance and scale

Check imaging distance and resolution in advance; include a scale with each specimen.

Overlapping images

Leave overlap between frames and take wider views where needed.

In the Yoshino study, adjacent images typically overlapped by one-half to two-thirds.

Yonenobu et al. (2026): Yoshino cedar

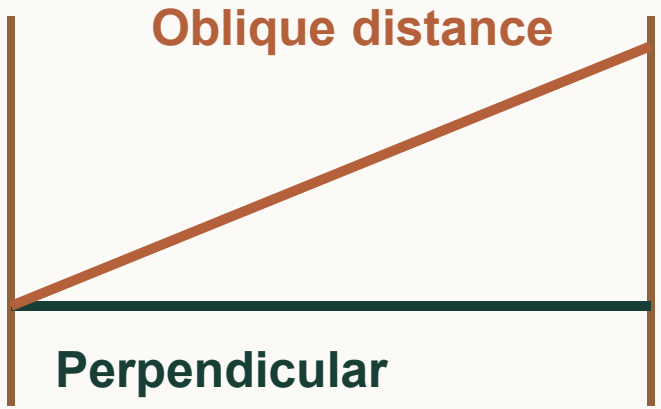
Yonenobu et al. (2026): koyamaki

The path relative to ring boundaries

An oblique path gives a longer distance between boundaries.

Adjust the path to the local orientation of the boundaries.

Preserve the measured path in images and records.



Same pair of boundaries

Synthetic teaching data / schematic example, not empirical tree measurements.

Convert pixels to a physical length

A known 10 mm spans 600 pixels in the image.

Spatial scale: 600 ÷ 10 = 60 px/mm

Width (mm) = boundary separation in pixels ÷ 60

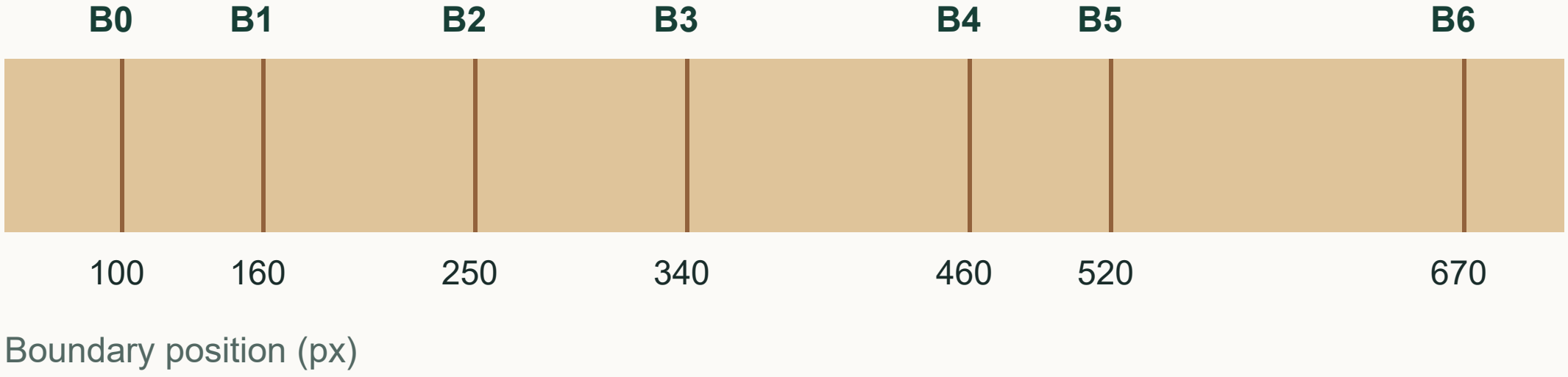
Check that the scale applies to the image and transformation.

From boundary positions to ring widths

Seven boundary positions define six widths.

Subtract successive positions, then apply the spatial scale.

Positions increase from inner to outer wood in this example.



Synthetic teaching data / schematic example, not empirical tree measurements.

Exercise A: calculate six ring widths

Boundary	Position (px)
B0	100
B1	160
B2	250
B3	340
B4	460
B5	520
B6	670

Scale: 600 px = 10 mm

Record each successive pixel difference and width in mm.

Compare the sum of six widths with the endpoint distance.

Synthetic teaching data / schematic example, not empirical tree measurements.

Exercise A: calculations and checks

Ring	Distance (px)	Width (mm)
1	60	1.00
2	90	1.50
3	90	1.50
4	120	2.00
5	60	1.00
6	150	2.50

600 ÷ 10 = 60 px/mm

(670 − 100) ÷ 60 = 9.50 mm

An endpoint check does not prove the boundaries were identified correctly.

Synthetic teaching data / schematic example, not empirical tree measurements.

R29 ImageJ documentation

Quality control links numbers and observations

Units, precision and order

Check pixels versus mm, orientation, duplicate or missing rows and delimiters.

Suspicious values

Investigate extremes, jumps and repeated values in the original image.

Reasons for revision

Recheck boundaries or transcription; preserve the change and its reason.

Connect measurement with dating checks

Assign candidate calendar years

Match a dated reference and record overlap and mismatches.

Return to the specimen

Investigate local absences, false rings, missed boundaries or measurement shifts.

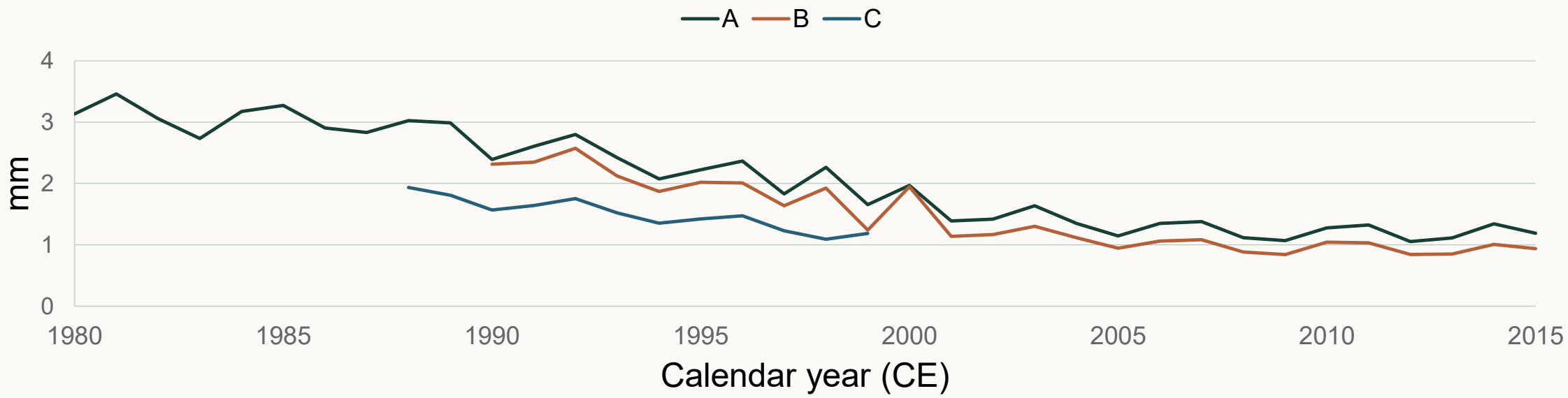
Separate observations and interpretation

Keep missing-position and dating hypotheses separate from observations.

Growth levels differ among trees

Width reflects species, size, individual history and other influences.

A raw mean alone need not isolate the signal of interest.



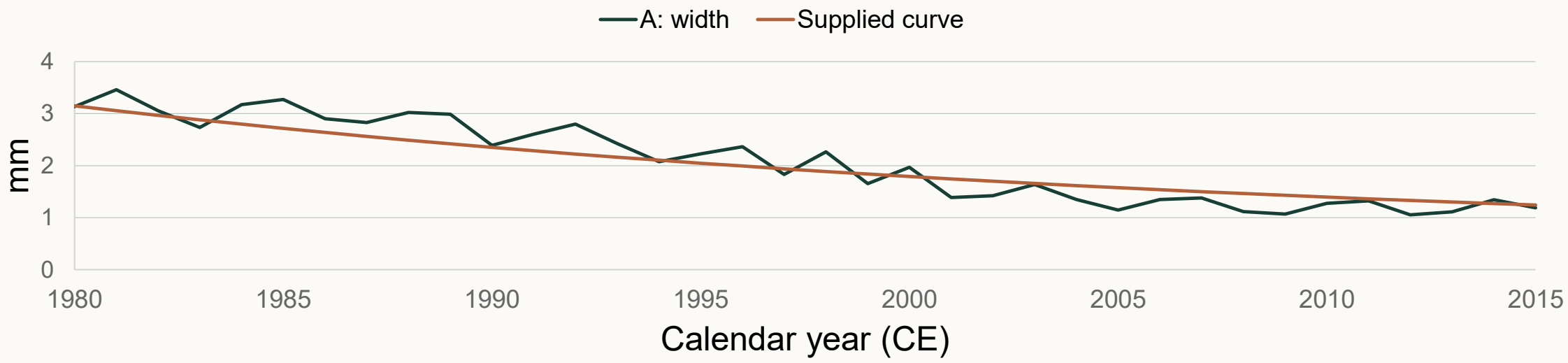
Synthetic teaching data / schematic example, not empirical tree measurements.

Standardization relative to a growth curve

Here, index = ring width ÷ supplied growth-curve value.

An index of 1 matches the curve; 1.2 is 1.2 times it.

The curve choice affects the variation retained.



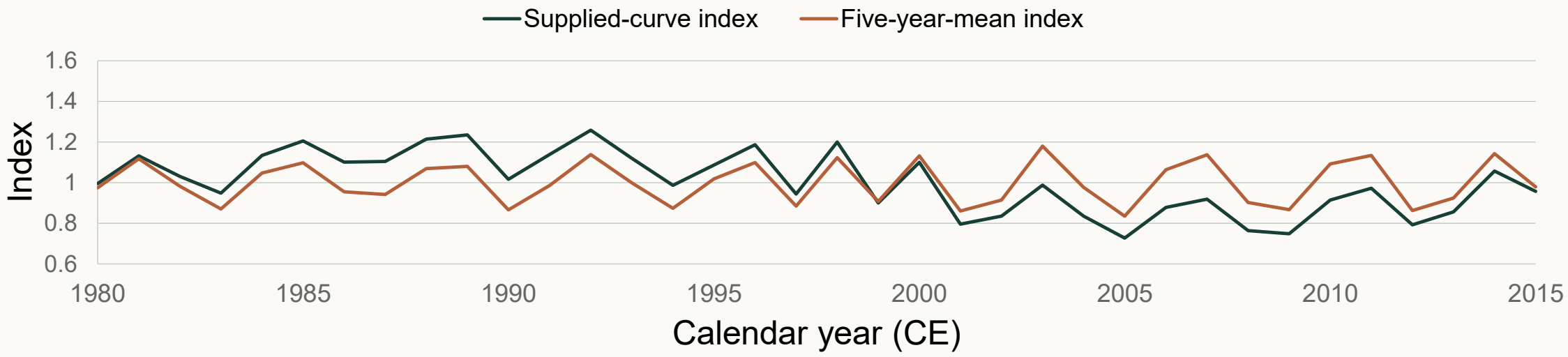
Synthetic teaching data / schematic example, not empirical tree measurements.

Which variation is retained?

A is indexed against two different reference curves.

A flexible reference can remove longer-term variation.

Record the question, method and settings; compare results.



Synthetic teaching data / schematic example, not empirical tree measurements.

Radii and the weighting of trees

Tree A has two radii: 0.8 and 1.2. Tree B has one: 1.5.

Average A first: 1.0. The equally weighted tree mean is 1.25.

A direct mean of all three radii is about 1.17, giving A more weight.

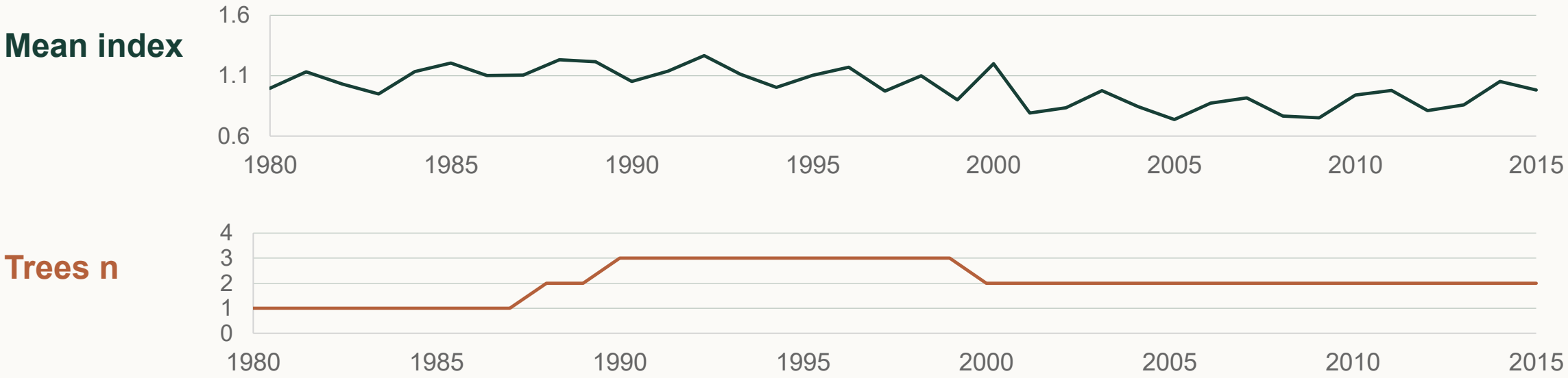
Synthetic teaching data / schematic example, not empirical tree measurements.

Show sample depth through time

Average the available tree indices in each calendar year.

A continuous chronology can have changing sample depth.

Tree count alone is insufficient; consider agreement and sampling bias.



Synthetic teaching data / schematic example, not empirical tree measurements.

Exercise B: build indices and a chronology

Calculate A, B and C indices from widths and supplied curves in the CSV.

Average the available indices by calendar year and count trees n.

Compare 1998–2000 and consider the effect of replacing blanks with zero.

Synthetic teaching data / schematic example, not empirical tree measurements.

What can the chronology establish?

An index is not temperature or precipitation

Test environmental responses against independent evidence.

A low index can have several causes

Distinguish drought, cold, disturbance and measurement problems.

Review processing for the question

Removed variation may matter when studying growth release.

Keep results traceable to the observations

01 Specimen ID, images, boundaries, units and calibration

02 Dating evidence and revision history

03 Detrending settings and sample depth

04 Preserve observations and store derived results separately.

Check methods and evidence in research papers

What to check

Specimen origins, methods, results and limitations.

Use a DOI

Add https://doi.org/ before the DOI and open the link.

Example: Yoshino cedar (Yonenobu et al., 2026)

https://doi.org/10.1016/j.dendro.2026.126572

Check full-text access; see the research-paper guide for further options.

Links for methods and further study

Sample preparation and measurement	USGS Dendrochronology Lab
Spatial calibration in ImageJ	ImageJ documentation
Tree-ring data and processing	NOAA NCEI
dplR: detrending and chronologies	dplR authors / CRAN
International Tree-Ring Data Bank	NOAA NCEI
Glossary of Dendrochronology	WSL

Exit questions

Does a matching total length prove that the boundaries are correct?

If C is blank in 2000, should it become zero in a three-tree mean?

Write a one-sentence reason for each answer.

Next: wood dates and human history

Lesson 5 introduces dendroarchaeology.

How does an outer-ring date relate to felling, working and construction?

Review surviving outer wood and evidence for the dating.